

Outputs and Quality Control

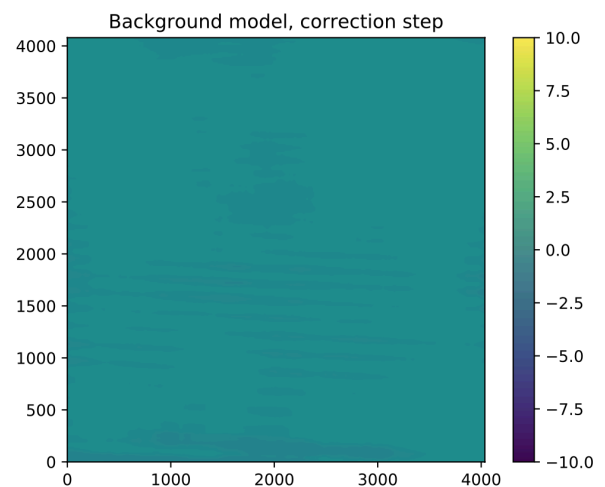
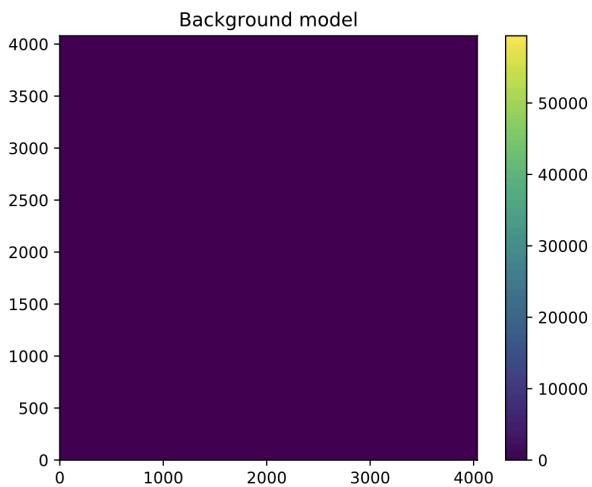
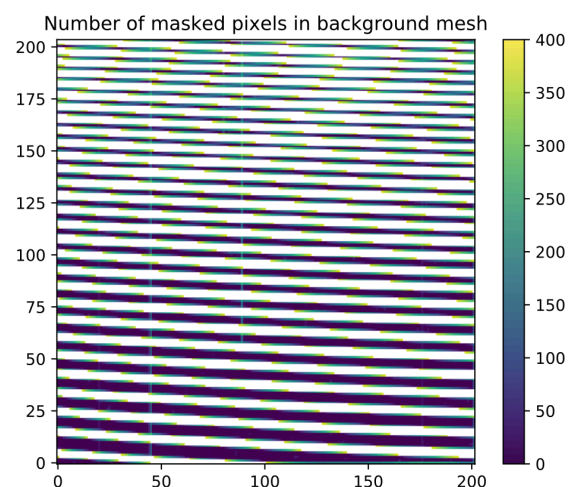
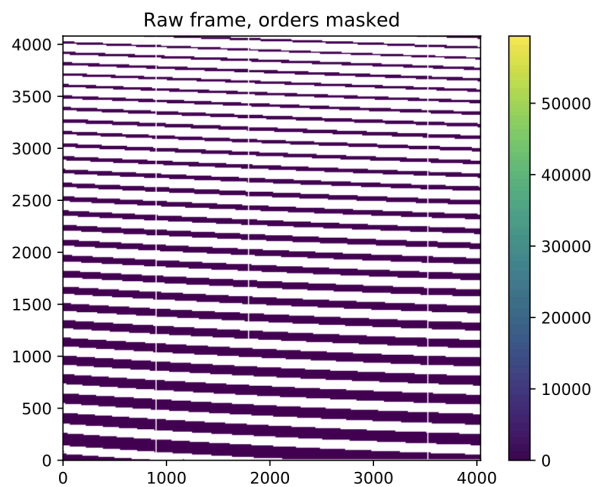
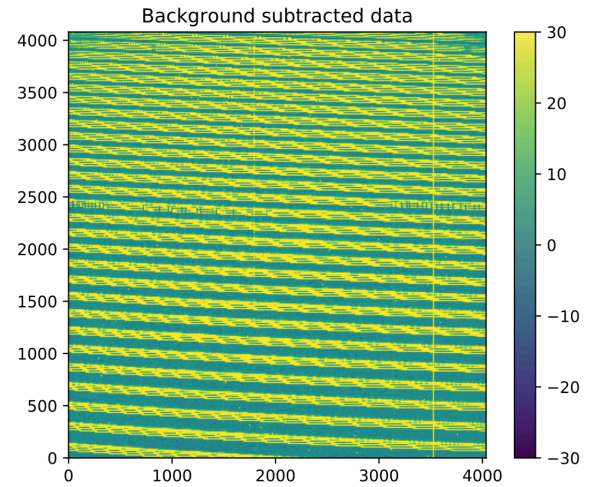
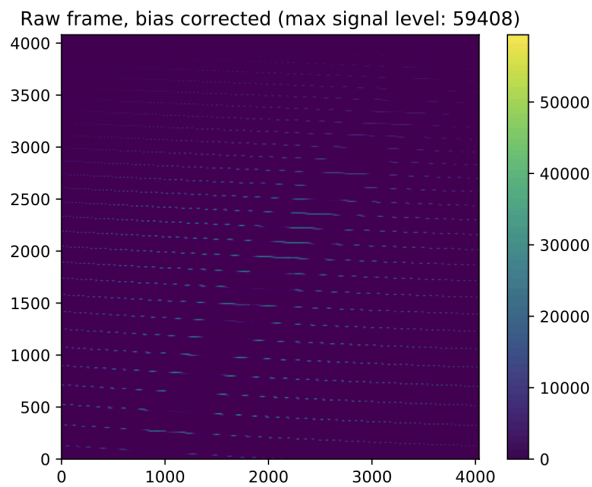
Created Jan. 2025

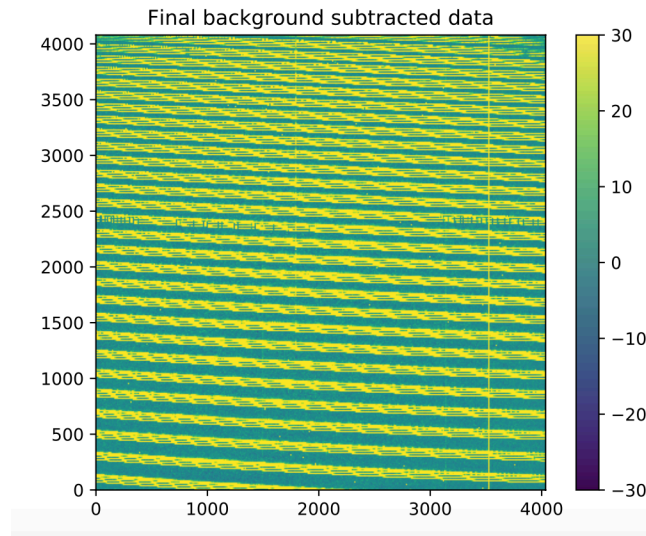
1. Reduction

a. Flux extractions

- i. Plots – ending ‘_backgroundfit.pdf’. This step is run on both science and flat frames.

Checking for any outliers – making sure there’s no clear orders that’ve been missed





ii. Logs – ending ‘.log’ (as in ‘SOOOE_r_o300.log’)

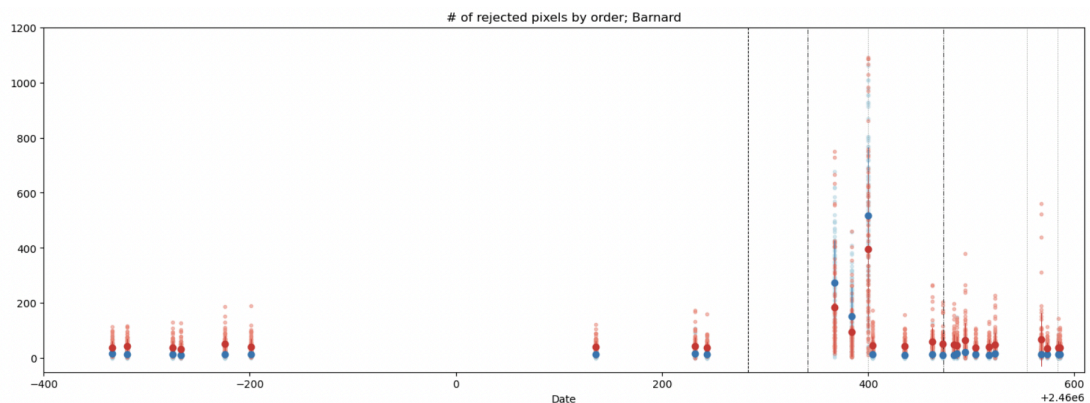
Reports darks and flats selected, and type of flux extraction performed (science has optimal extraction in fibers 2-4 and box in fiber 5). Also reports number and % of rejected pixels in fiber 2-4, per order and fiber. It'll also report any columns rejected due to an excessive number of rejected pixels. Lastly, it lists the SNR in each order.

Red flags here would be a high number of rejected pixels (anything > 100 pixels is worth inspection). As we discovered with HD3651, high signal targets may also have a large number of pixels rejected.

Here's an example of an ideal log:

```
2023-11-08 15:05:51,475 - 20231014T145138Z_SO00E_r_0600 - INFO - Optimum extraction in fiber_4, order 72
2023-11-08 15:05:56,783 - 20231014T145138Z_SO00E_r_0600 - WARNING - Too many bad pixels in column 125, reject column
2023-11-08 15:05:57,285 - 20231014T145138Z_SO00E_r_0600 - INFO - Rejected 108 pixels in 60 columns during optimal extraction.
2023-11-08 15:05:57,286 - 20231014T145138Z_SO00E_r_0600 - INFO - Rejections with flux changes < 0.5%: 0 (0%)
2023-11-08 15:05:57,286 - 20231014T145138Z_SO00E_r_0600 - INFO - Optimum extraction in fiber_4, order 73
2023-11-08 15:06:02,873 - 20231014T145138Z_SO00E_r_0600 - INFO - Rejected 82 pixels in 32 columns during optimal extraction.
2023-11-08 15:06:02,873 - 20231014T145138Z_SO00E_r_0600 - INFO - Rejections with flux changes < 0.5%: 0 (0%)
2023-11-08 15:06:02,873 - 20231014T145138Z_SO00E_r_0600 - INFO - Optimum extraction in fiber_4, order 74
2023-11-08 15:06:08,494 - 20231014T145138Z_SO00E_r_0600 - INFO - Rejected 77 pixels in 35 columns during optimal extraction.
2023-11-08 15:06:08,494 - 20231014T145138Z_SO00E_r_0600 - INFO - Rejections with flux changes < 0.5%: 0 (0%)
2023-11-08 15:06:08,494 - 20231014T145138Z_SO00E_r_0600 - INFO - Optimum extraction in fiber_4, order 75
2023-11-08 15:06:14,065 - 20231014T145138Z_SO00E_r_0600 - INFO - Rejected 77 pixels in 28 columns during optimal extraction.
2023-11-08 15:06:14,066 - 20231014T145138Z_SO00E_r_0600 - INFO - Rejections with flux changes < 0.5%: 0 (0%)
```

And how the number of rejected pixels will increase with poor flats (..300-..400 JD):



b. Etalon line-fitting

i. Logs – ending ‘etalon_peakfit.log’

Reports ... a lot of information. The biggest red flags are, not surprisingly, warnings and error messages. Error messages in science frame fittings should *always* be investigated. The most common I’ve seen is from a mismatch in the number of peak maxima and minima. This is usually caused by leftover image artifacts or cosmics, and can be solved by calling an etalon frame as a “guiding example”.

DLLLE/L log errors are ok as long as they’re for blue frames order >117 for fibers 2-4(5), OR for peaks ~906 and ~1787 for red frames. Order 90 errors for blue DEEEE frames aren’t too concerning either.

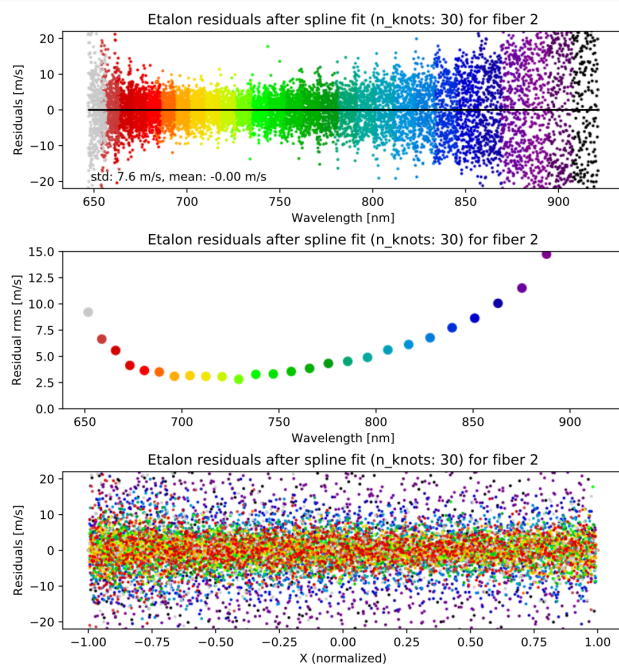
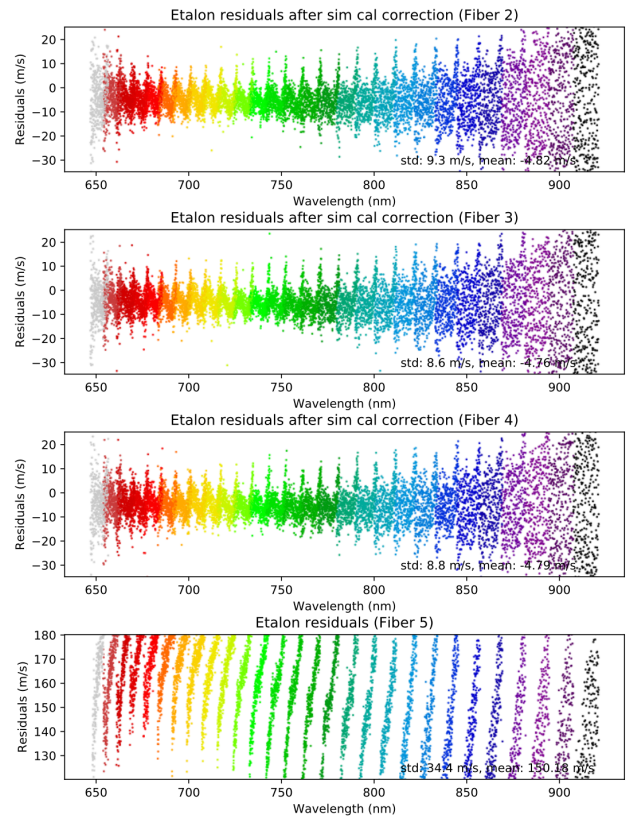
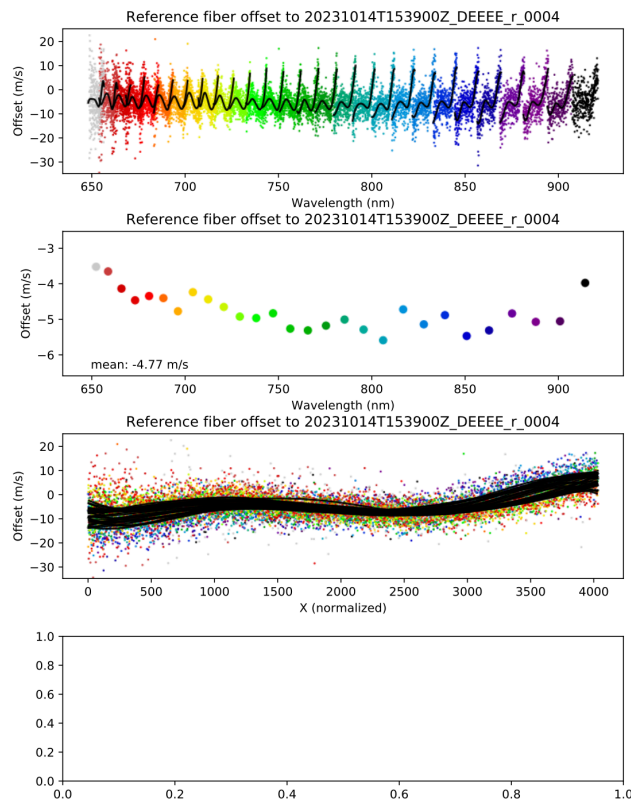
Here’s an example of the type of information reported in these logs:

```
2023-12-20 22:15:31,082 - etalon_positions.fit - INFO - Fitting 5_122 done in 32.40s
2023-12-20 22:15:31,170 - etalon_positions.fit - INFO - Fit residuals on 5_99 (1. iteration): 224.4 (0.059)
2023-12-20 22:15:31,188 - etalon_positions.fit - INFO - Slit half-width on 5_123 (2. iteration): 1.329 - 2.073
2023-12-20 22:15:31,189 - etalon_positions.fit - INFO - Flat-relative amplitudes on 5_123 (2. iteration): min: 0.03, max: 0.04, median: 0.04
2023-12-20 22:15:31,199 - etalon_positions.fit - INFO - Mean shift in linecenters for 5_101 (2. vs. 1. iteration): 0.29 m/s
2023-12-20 22:15:31,200 - etalon_positions.fit - INFO - Bulk shift in linecenters for 5_101 (2. vs. 1. iteration): -0.12 m/s
2023-12-20 22:15:31,244 - etalon_positions.fit - INFO - Fit residuals on 5_104 (2. iteration): 204.8 (0.054)
2023-12-20 22:15:31,305 - etalon_positions.fit - INFO - Slit half-width on 5_120 (2. iteration): 1.312 - 2.059
2023-12-20 22:15:31,306 - etalon_positions.fit - INFO - Flat-relative amplitudes on 5_120 (2. iteration): min: 0.02, max: 0.03, median: 0.03
2023-12-20 22:15:31,568 - etalon_positions.fit - INFO - Slit half-width on 5_124 (2. iteration): 1.371 - 2.074
2023-12-20 22:15:31,569 - etalon_positions.fit - INFO - Flat-relative amplitudes on 5_124 (2. iteration): min: 0.04, max: 0.06, median: 0.04
2023-12-20 22:15:31,640 - etalon_positions.fit - INFO - Fit residuals on 5_101 (2. iteration): 199.2 (0.053)
2023-12-20 22:15:31,677 - etalon_positions.fit - INFO - Slit half-width on 5_98 (2. iteration): 1.323 - 2.027
2023-12-20 22:15:31,678 - etalon_positions.fit - INFO - Flat-relative amplitudes on 5_98 (2. iteration): min: 0.08, max: 0.09, median: 0.08
2023-12-20 22:15:32,298 - etalon_positions.fit - INFO - Slit half-width on 5_92 (2. iteration): 1.299 - 2.019
2023-12-20 22:15:32,299 - etalon_positions.fit - INFO - Flat-relative amplitudes on 5_92 (2. iteration): min: 0.14, max: 0.16, median: 0.15
2023-12-20 22:15:32,573 - etalon_positions.fit - INFO - Slit half-width on 5_117 (2. iteration): 1.335 - 2.047
2023-12-20 22:15:32,573 - etalon_positions.fit - INFO - Flat-relative amplitudes on 5_117 (2. iteration): min: 0.03, max: 0.06, median: 0.04
2023-12-20 22:15:32,732 - etalon_positions.fit - INFO - Slit half-width on 5_94 (2. iteration): 1.311 - 2.024
2023-12-20 22:15:32,742 - etalon_positions.fit - INFO - Flat-relative amplitudes on 5_94 (2. iteration): min: 0.11, max: 0.14, median: 0.13
2023-12-20 22:15:32,774 - etalon_positions.fit - INFO - Mean shift in linecenters for 5_119 (2. vs. 1. iteration): 0.18 m/s
2023-12-20 22:15:32,774 - etalon_positions.fit - INFO - Bulk shift in linecenters for 5_119 (2. vs. 1. iteration): -0.01 m/s
2023-12-20 22:15:33,132 - etalon_positions.fit - INFO - Fit residuals on 5_119 (2. iteration): 183.0 (0.048)
2023-12-20 22:15:33,146 - etalon_positions.fit - INFO - No more improvement for 5_119 after 2. iteration, finished
```

c. Spline/wavelength solution

i. Plots – ending ‘_spline.pdf’

Reports offset per order between SOOOE and DEEEE calibration frames. Also shows residuals in each fiber before and after correcting for the etalon wavelength solution.



It reports the left 3 plots for fibers 2-5. Red flags would be extremely high residuals, or orders that don't follow the curved trend.

ii. Logs – ending ‘_spline.log’

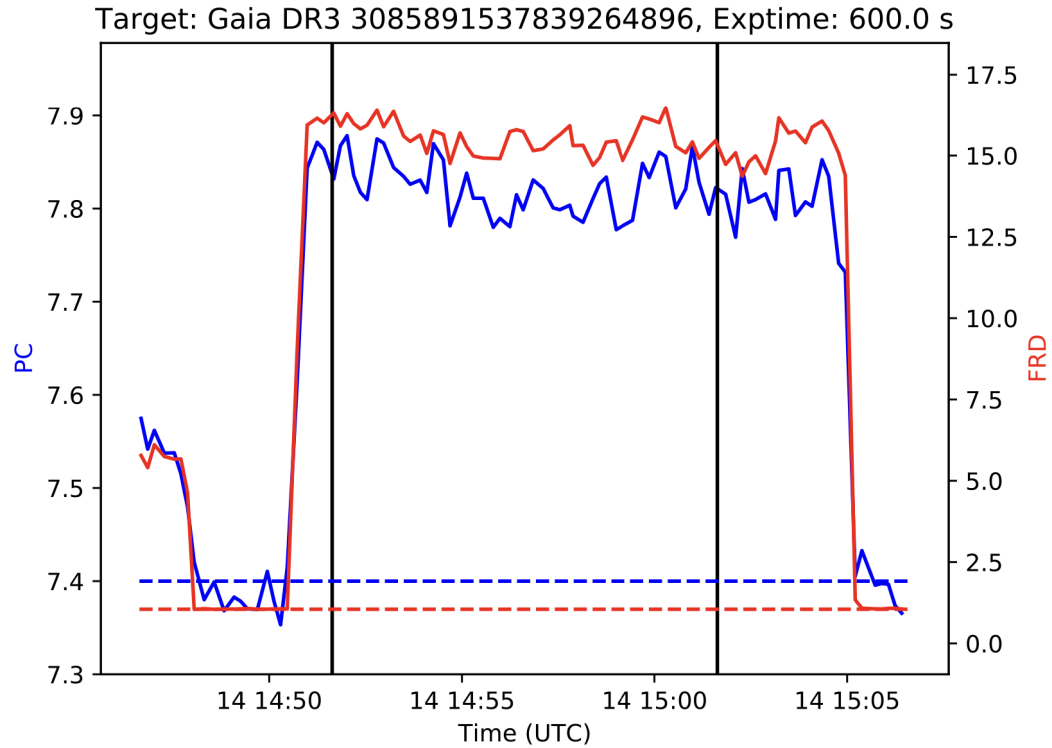
Reports pixel offset for fiber 5 per order (offset in the SOOOE frame relative to the nearest etalon frame). Reports bad lines found per order. Lastly, reports:

- the instrument drift;
- relative drift in fiber 5;
- relative drift in fiber 5 based on the spline;
- relative drifts in fiber 2-4;
- and absolute drifts in fibers 2-4 in the etalon frames.

d. Barycentric Corrections

i. Plots – ending ‘_exposuremeter.pdf’

Looking to make sure that the zeropoints line up to the actual flux – as well as any anomalies in the flux (sudden drop or spike, especially a drop below the zeropoint during the exposure)



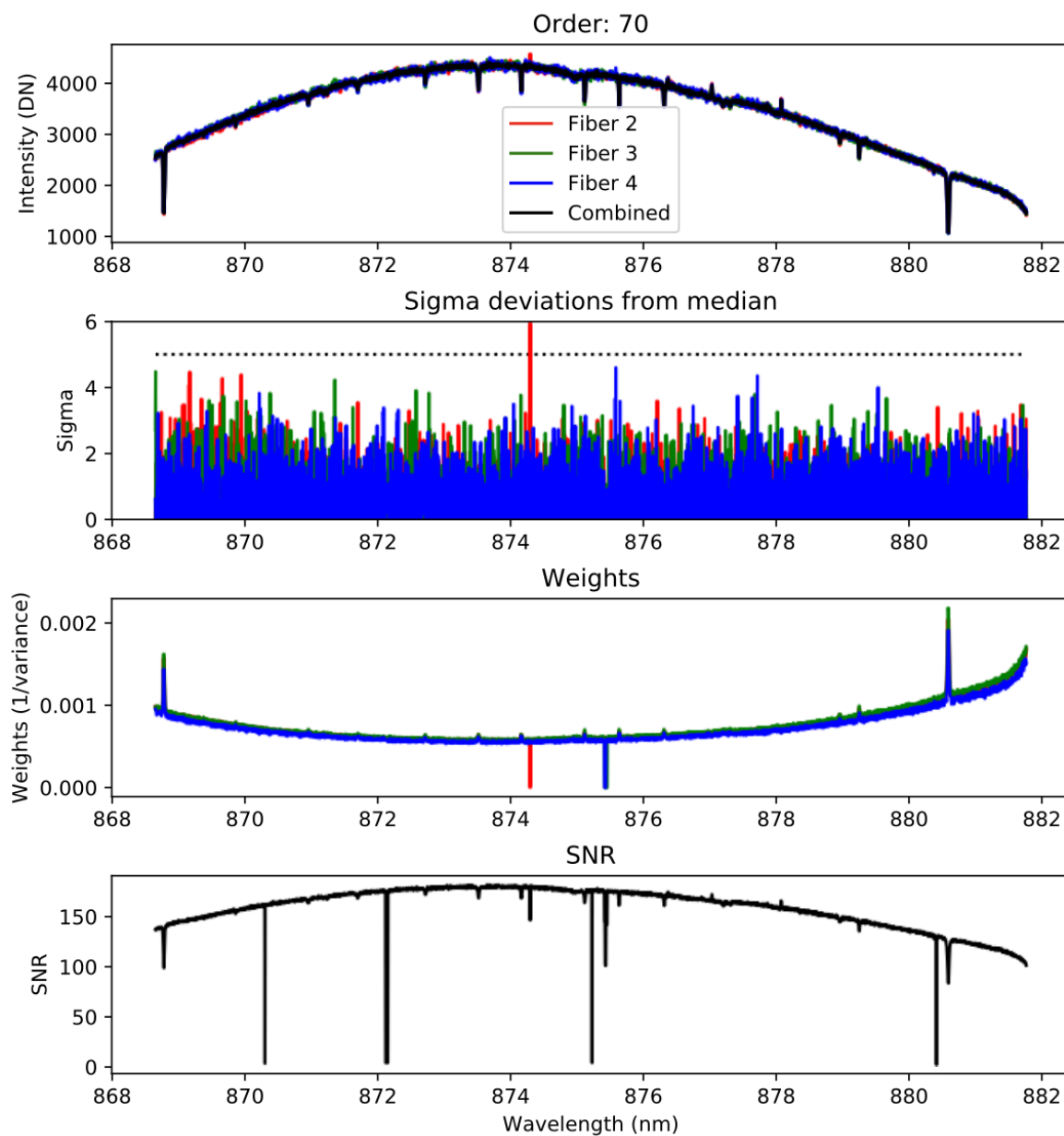
ii. Logs – ending ‘exposuremeter.log’

Reports exposure start time, midpoint time, and fluxweighted midpoints. Reports barycentric Earth RV at nominal and fluxweighted midpoints, as well as BERV DV/DT. Lastly, reports zeropoints for FRD and PC channels, as well as min, max, median, and std for each, plus the scalefactor of med FRD/PC.

e. Fiber 6**i. Plots – ending ‘_fiber6.pdf’**

Looking to make sure that there's no clear outliers (that all fibers reasonably agree).

Saves the below plot for each order:



ii. Logs – ending ‘_fiber6.log’

Reports number of clipped pixels, number of NaNs, masked pixels, and kappa_sigma per order. Clipped pixels should be under 20, and should report a warning if that number is exceeded.

An example of a good log:

```
2024-01-02 13:50:32,206 - fibercombiner - INFO - Kappa_sigma in order 85: 5.0
2024-01-02 13:50:32,207 - fibercombiner - INFO - Clipped 0 pixels in fiber 2 of order 85
2024-01-02 13:50:32,207 - fibercombiner - INFO - Clipped 0 pixels in fiber 3 of order 85
2024-01-02 13:50:32,207 - fibercombiner - INFO - Clipped 1 pixels in fiber 4 of order 85
2024-01-02 13:50:32,223 - fibercombiner - INFO - Number of NaNs fixed in order 85: 19
2024-01-02 13:50:32,883 - fibercombiner - INFO - Mask additional pixels in red arm at pixel 1793 and 1794
2024-01-02 13:50:32,892 - fibercombiner - INFO - Kappa_sigma in order 86: 5.0
2024-01-02 13:50:32,893 - fibercombiner - INFO - Clipped 0 pixels in fiber 2 of order 86
2024-01-02 13:50:32,893 - fibercombiner - INFO - Clipped 0 pixels in fiber 3 of order 86
2024-01-02 13:50:32,893 - fibercombiner - INFO - Clipped 1 pixels in fiber 4 of order 86
2024-01-02 13:50:32,910 - fibercombiner - INFO - Number of NaNs fixed in order 86: 19
2024-01-02 13:50:33,586 - fibercombiner - INFO - Mask additional pixels in red arm at pixel 1793 and 1794
2024-01-02 13:50:33,594 - fibercombiner - INFO - Kappa_sigma in order 87: 5.0
2024-01-02 13:50:33,595 - fibercombiner - INFO - Clipped 1 pixels in fiber 2 of order 87
2024-01-02 13:50:33,595 - fibercombiner - INFO - Clipped 1 pixels in fiber 3 of order 87
2024-01-02 13:50:33,595 - fibercombiner - INFO - Clipped 0 pixels in fiber 4 of order 87
2024-01-02 13:50:33,611 - fibercombiner - INFO - Number of NaNs fixed in order 87: 21
2024-01-02 13:50:34,252 - fibercombiner - INFO - Mask additional pixels in red arm at pixel 1793 and 1794
```

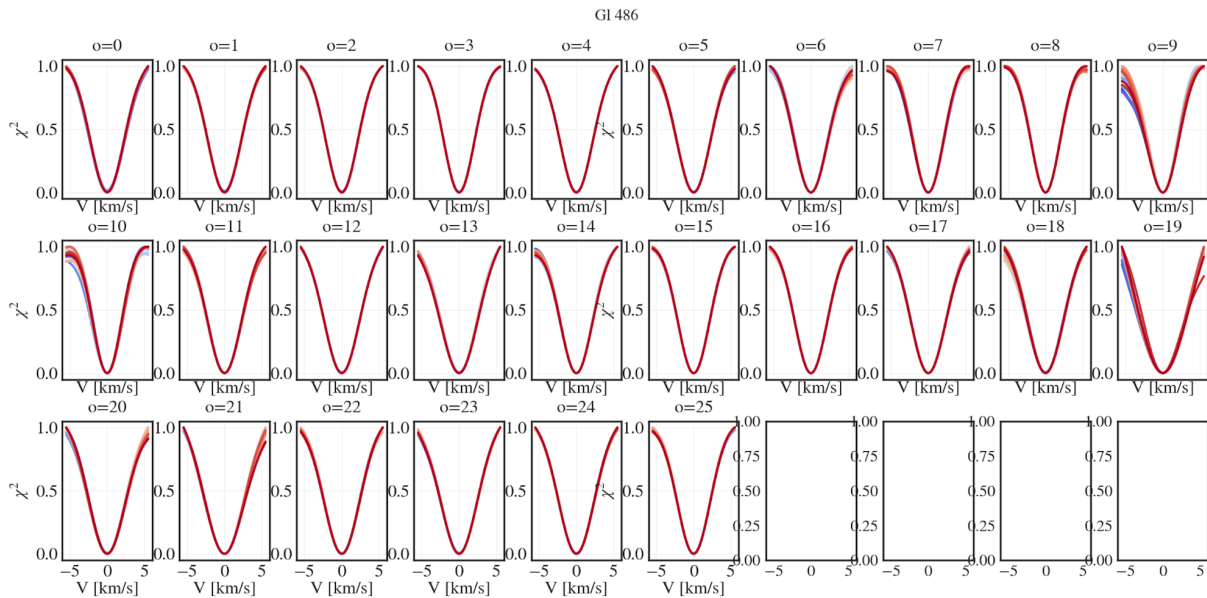
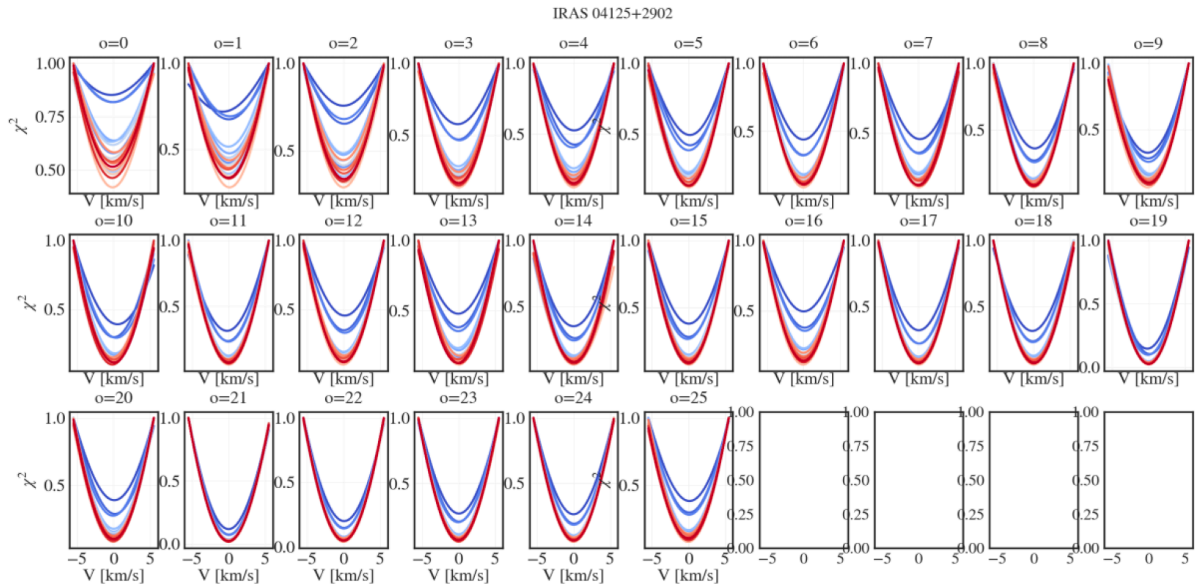
An example of a concerning log (though it doesn't necessarily mean there's an issue with the data):

```
2022-09-07 12:48:20,690 - fibercombiner - INFO - Mask additional pixels in red arm at pixel 1793 and 1794
2022-09-07 12:48:20,699 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,700 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,701 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,703 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,704 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,705 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,706 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,708 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,709 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,710 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 67
2022-09-07 12:48:20,711 - fibercombiner - INFO - Kappa_sigma in order 67: 10.0
2022-09-07 12:48:20,712 - fibercombiner - INFO - Clipped 40 pixels in fiber 2 of order 67
2022-09-07 12:48:20,712 - fibercombiner - INFO - Clipped 0 pixels in fiber 3 of order 67
2022-09-07 12:48:20,712 - fibercombiner - INFO - Clipped 0 pixels in fiber 4 of order 67
2022-09-07 12:48:20,729 - fibercombiner - INFO - Number of NaNs fixed in order 67: 13
2022-09-07 12:48:21,381 - fibercombiner - INFO - Mask additional pixels in red arm at pixel 1793 and 1794
2022-09-07 12:48:21,389 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,390 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,392 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,393 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,394 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,395 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,397 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,398 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,399 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,400 - fibercombiner - WARNING - Number of maximum clipped pixels exceeded in order 68
2022-09-07 12:48:21,402 - fibercombiner - INFO - Kappa_sigma in order 68: 10.0
2022-09-07 12:48:21,402 - fibercombiner - INFO - Clipped 40 pixels in fiber 2 of order 68
2022-09-07 12:48:21,402 - fibercombiner - INFO - Clipped 2 pixels in fiber 3 of order 68
2022-09-07 12:48:21,402 - fibercombiner - INFO - Clipped 0 pixels in fiber 4 of order 68
2022-09-07 12:48:21,420 - fibercombiner - INFO - Number of NaNs fixed in order 68: 13
```


2. SERVAl

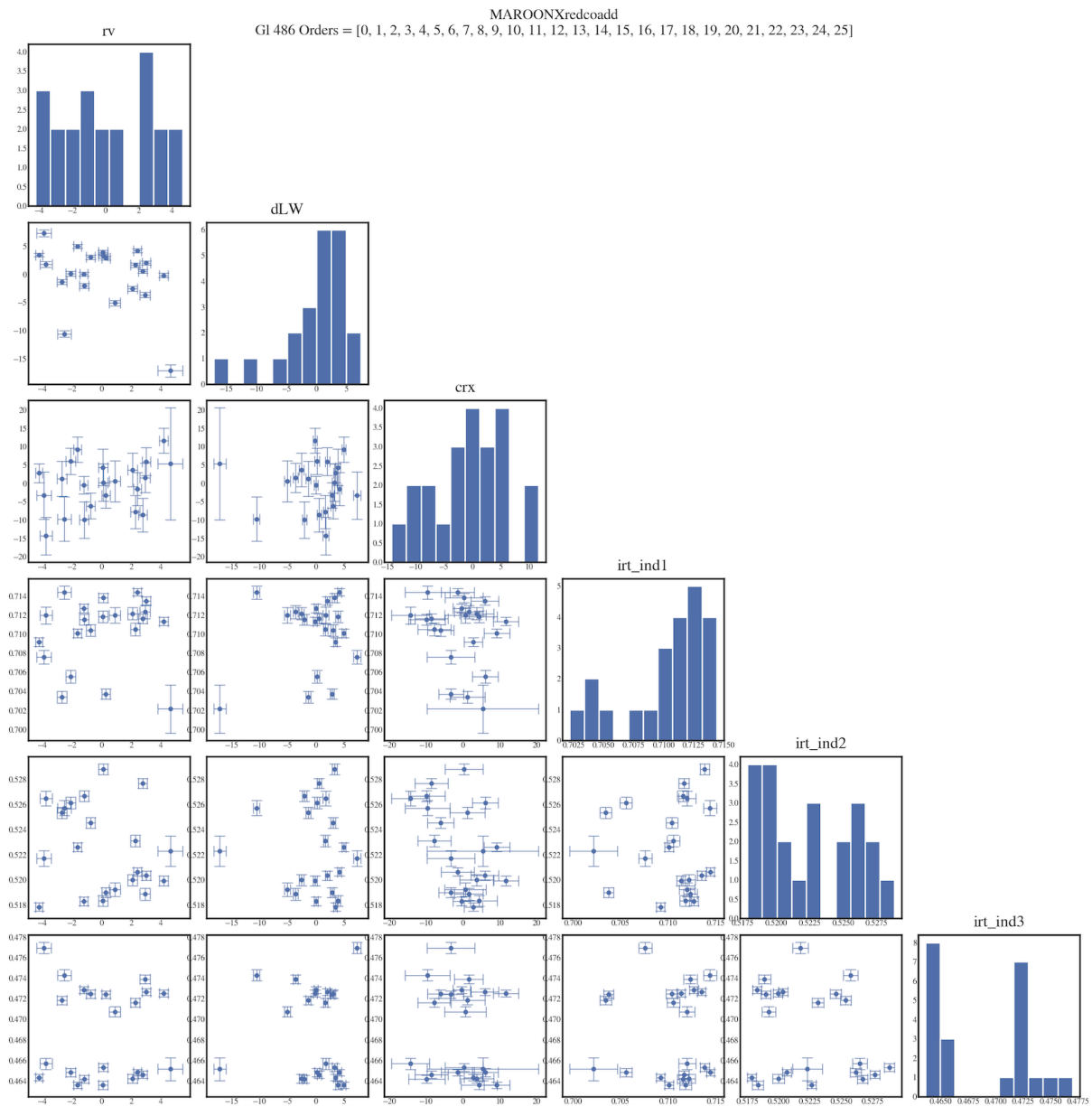
a. Chi2 – ‘_chi2map.png’

Looking for any outliers, especially anything off-center. Top image indicates ~ 3 problematic observations – the three dark blue lines. The bottom plot is ideal.



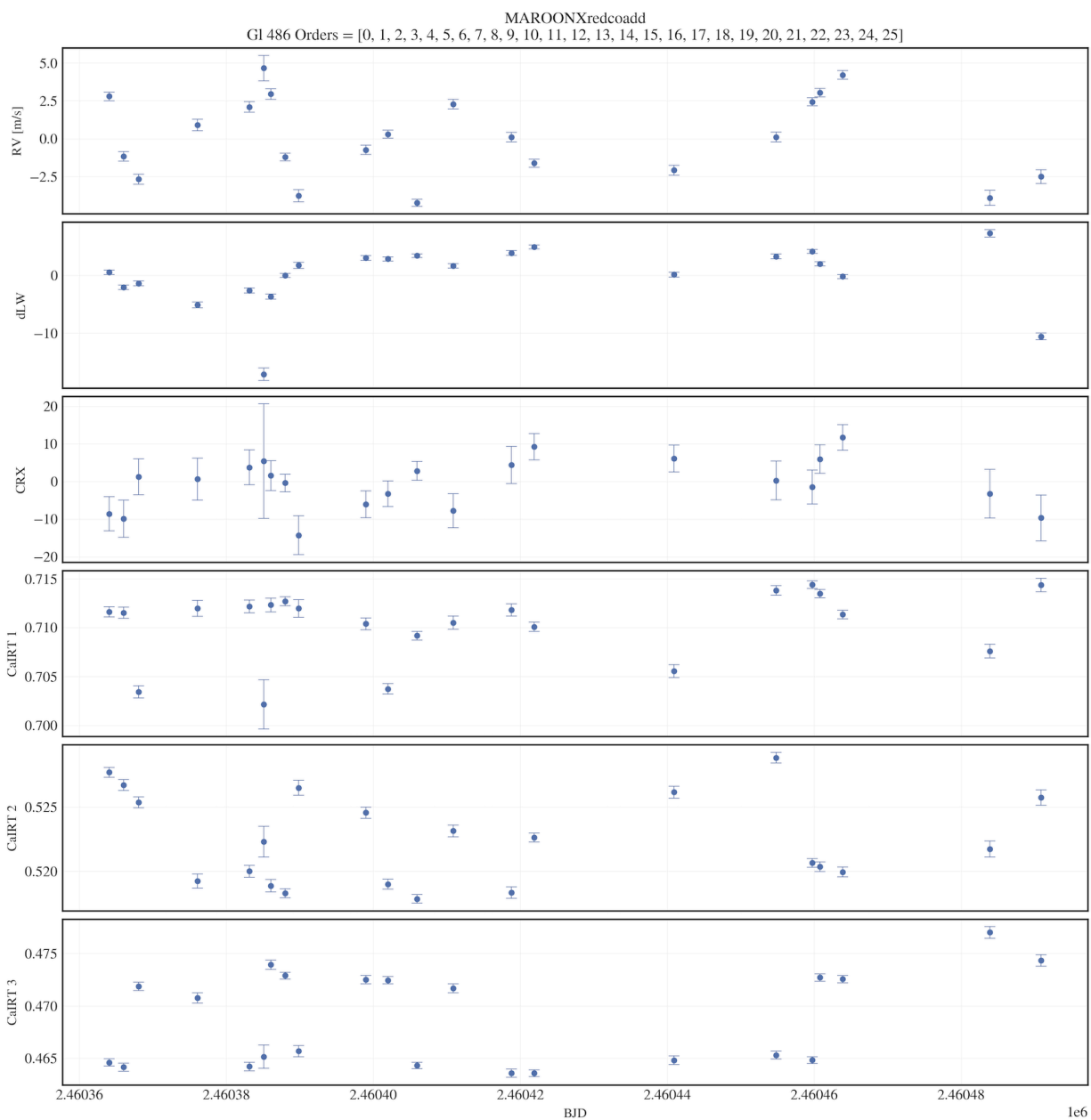
b. Correlation plots

There's several correlation plots (ending '`_correlation_rv_vs_[...].png`'), and an overview plot (ending '`_correlation_corner.png`'). Red flags would be any correlations that aren't easily explained by stellar activity.



c. RV activity panel

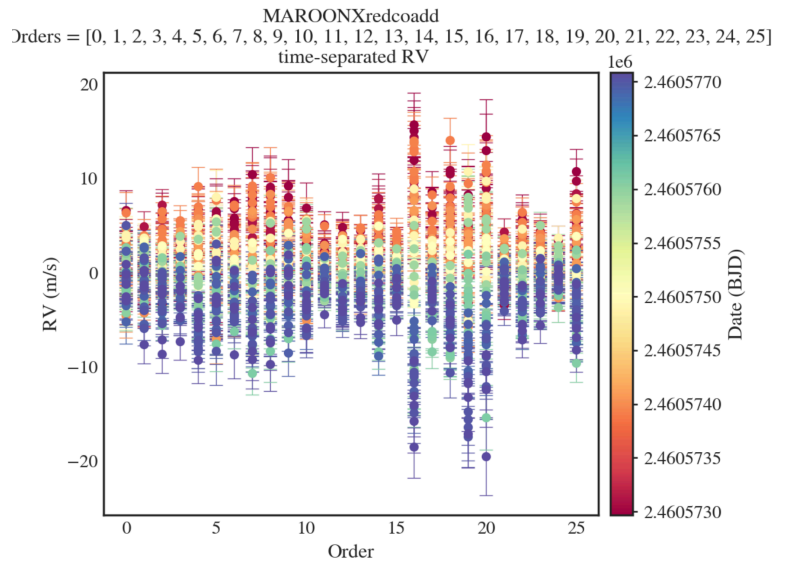
Ending ‘_rv_activity_panel.png’. Look for any trends over time.



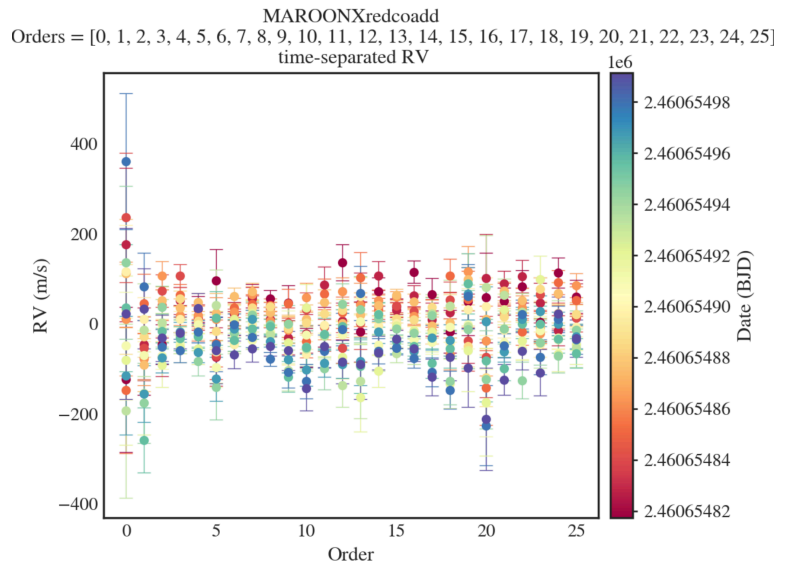
d. Order vs RV

Ending ‘_order_wl_RV.png’. Look for any strong outliers or trends. Similar plot is ‘_rv_vs_wave.png’, which plots RVs against wavelength.

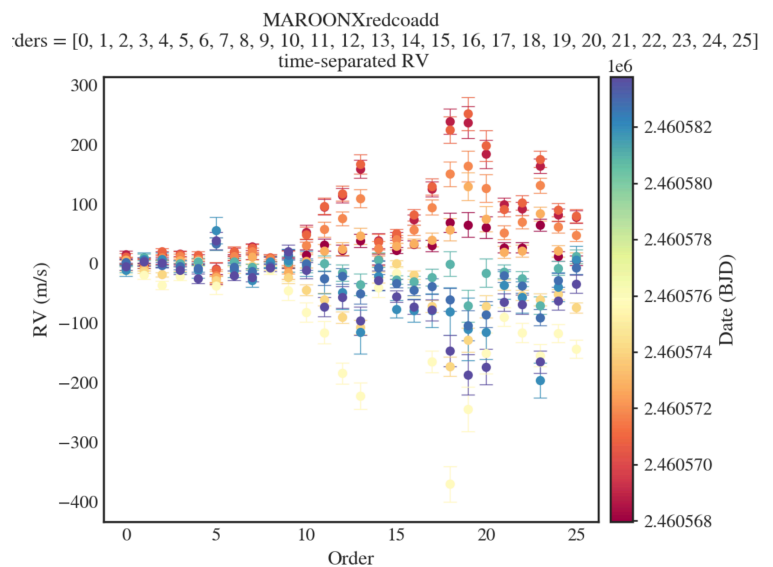
Top plot shows a strong RV slope over time, and is pretty consistent per order.



Middle plot indicates that the first order is poor.

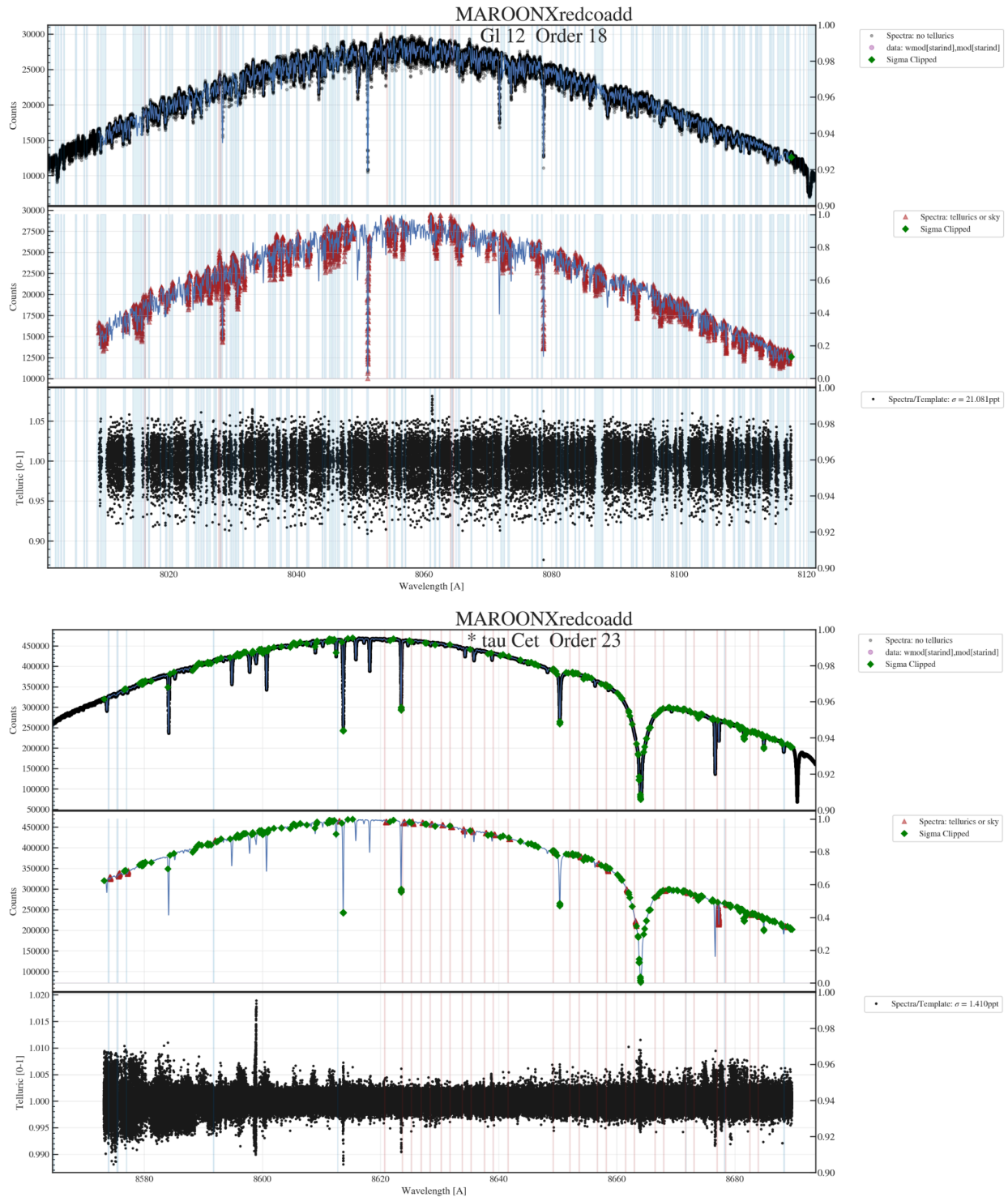


Last plot shows a discrepancy between first and second half of orders.



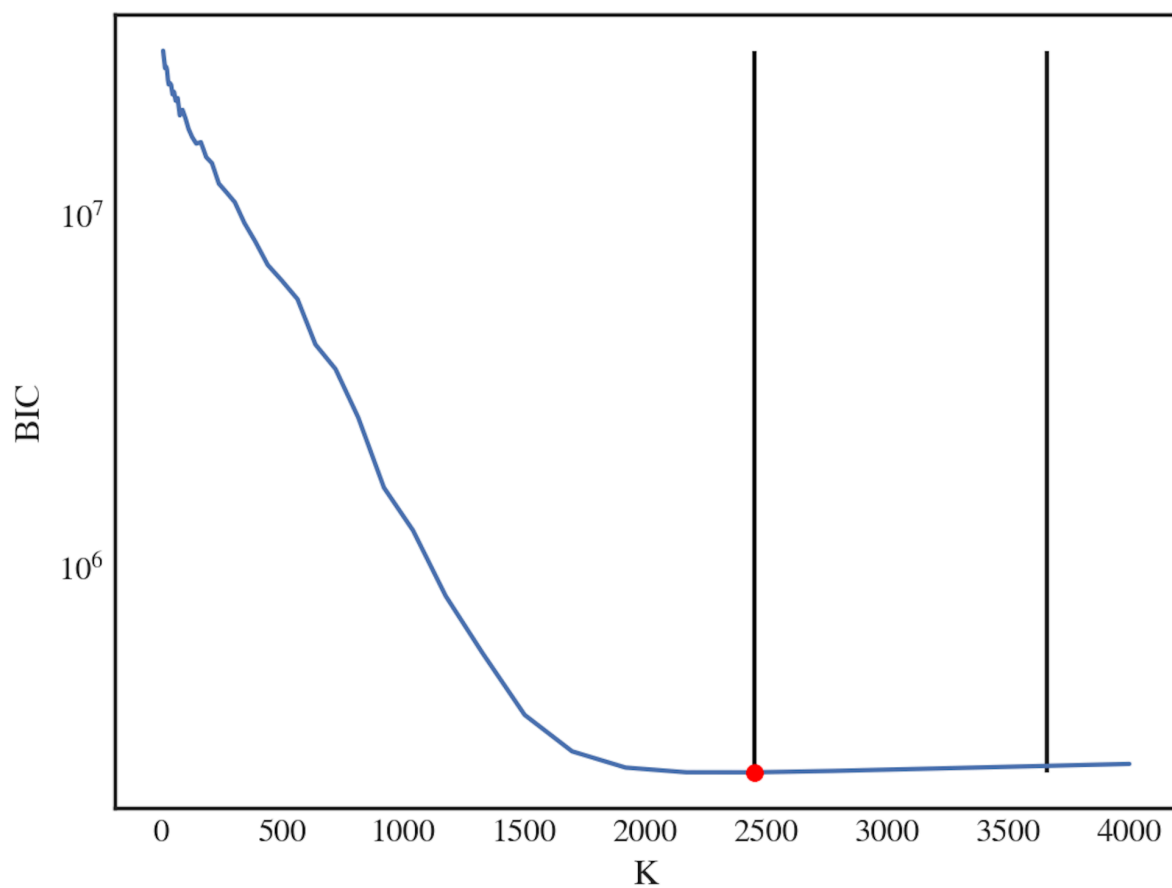
e. Template

Ending ‘_template_o_#.png’. Red flags would be high levels of sigma clipped points, or an unsteady/unorderly telluric residuals line. Top plot looks nice, bottom plot has way too many sigma clipped points at important features.



f. Template knots

Ending ‘_template_knots_o#.png’. Make sure that the red point is at a “minimum” and before the maximum K (at ~3500).



g. Stellar activity plots

In `‘/plots_stellar_activity’`, ending `‘____#_Halpha/ca_irt.png’`. Make sure that the spectra is aligned, showing that the systemic RV is correct. Also an indicator of spectral noise and stellar activity.

